

Scalability & Future Plan

Date	10 July 2026
Team ID	XXXXXX
Project Name	Raising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Description	Impact	Priority to Address (High / Medium / Low)
1	Missing real-time IoT sensor API.	Direct, automated rainfall and cloud data feeds from localized weather stations are not yet integrated.	Higher latency in predictive alerts; currently requires manual data entry.	High
2	Single core model optimization.	The localized .pkl model runs on a single thread.	Reduced processing speed under simultaneous queries; potential performance bottleneck.	Medium
3	Static risk thresholds.	Flood risk levels (Green/Yellow/Red) use static, non-adaptive meteorological parameters.	Risk levels may not accurately reflect rapid micro-climate shifts.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Minimal (system configured for localized developer testing).	Migrate Flask app to a containerized multi-node cluster with load balancers.
2	Data Storage	Local CSV file for prediction logging.	Implement a distributed relational database (e.g., PostgreSQL) for robust historical tracking.
3	Performance	Pre-processing of input vectors is basic and sequential.	Introduce asynchronous data ingestion pipelines using a task queue (e.g., Celery).
4	Security	No user authentication on the monitoring control panel.	Implement Role-Based Access Control (RBAC) and data encryption.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Complete the data ingestion automation layer with API mappings for localized weather APIs.	Q1 2027	Critical reduction in alert latency

Phase 2	(Completed data ingestion automation layer)	Q1 2027	Critical reduction in alert latency
Phase 3	Integrate Time-Series predictive capability using RNNs.	Q3 2027	Move from purely reactive prediction to anticipatory event forecasting.
Phase 4	Deploy full-scale municipal sensor network.	2028	Achieve high-density regional disaster resilience and localized safety guarantees.